rsna-knee-abnormality-detection

10th-Place Solution

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Team	MEJAL
Author	LP
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10th-Place Solution: Multi-Source Soft Labels and Multi-Sequence Knee MRI Ensembles

We thank RSNA, Kaggle, the institutions that provided the images and reports, the expert annotators, and the participants who shared models and code. We also thank our teammates for their collaboration on labels, imaging models, the CoAtNet branch, and the final ensemble.

Final team result: Public LB 0.967, Private LB 0.957, 10th place. The final solution combined a multi-backbone, multi-sequence model group, a CoAtNet-2 model group, and public R7 models.

1. Team Overview: Two Development Paths, One Imaging Task

The competition required predicting **12 abnormalities** from a knee MRI examination: ACL, MCL, medial and lateral meniscus abnormalities, osteoarthritis in three compartments, effusion, synovitis, Baker's cyst, contusion, and fracture. The training set contained **4,407 studies**: 4,349 with images and radiology reports, and 58 with official expert labels. Reports were unavailable at test time. The metric was the unweighted mean of the 12 class-wise ROC AUCs, making **the ranking of studies within each class** central to the score. Competition overview

Our solution had three main components: training targets that better matched the competition's positive-label definitions; aggregation of spatial, slice, and sequence evidence from MRI; and an ensemble of prediction branches with different supervision and architectures.

The team developed two main paths. The multi-backbone, multi-sequence path emphasized supervision alignment and hierarchical evidence aggregation. The CoAtNet path emphasized iterative label distillation and window-level multiple-instance learning (MIL). We describe them separately before explaining their later convergence and final ensemble.

Development path	Core design
Multi-source soft supervision and multi-sequence evidence aggregation	LLM/VLM label blending, OAI label mapping, spatial/slice/sequence aggregation, and validation across backbones.
Iterative label distillation and CoAtNet MIL	Image-only readings, iterative out-of-fold labels, window-level attention MIL, full-data seeds, and all-window fine-tuning.

These are development and implementation boundaries, not a claim that the final supervision sources were independent. Labels and predictions from both paths were combined during the later team collaboration, as described in Section 4.

2. Path One: Multi-Source Soft Supervision and Multi-Sequence Evidence Aggregation

This path was developed jointly by Evan zhu, LP, minmin andy, and Jak ma.

Evan zhu primarily drove prompt design, supervision development, and overall technical planning. Evan refined training supervision through report interpretation, image–report integration, and OAI label mapping, aligned label definitions with the competition targets, and proposed key ideas behind the final-stage score improvements.

LP refined the overall technical approach and worked on data-quality analysis and processing optimization. Key contributions included building multi-sequence visual models, core module design and optimization, complementary training across heterogeneous label sources, and improvements to multi-stage iterative distillation strategies. LP carried out model training and led model tuning, ablation studies, and ensembling, assisted with post-processing improvements, and worked on integrating the team's approaches and final solution decisions.

minmin andy and Jak ma jointly contributed to model training and iterative experiments across multiple backbones. They ran experiments with different backbone and training configurations, assisted with model tuning and result validation, and worked on inference-pipeline optimization and speed improvements, supporting candidate-model preparation, continued refinement, and the final solution.

Our multi-backbone branch centered on one question: **when inference uses only MRI but most training supervision comes from reports, how can we better align the learning targets and visible evidence with the competition task?** An unmentioned finding is not necessarily image-negative, and a reported abnormality may not meet the competition's positive criteria. Evidence may also appear only in a local region, a few slices, or one sequence. Supervision determines what the model learns; architecture determines how it organizes evidence. Their value must ultimately be established through downstream experiments.

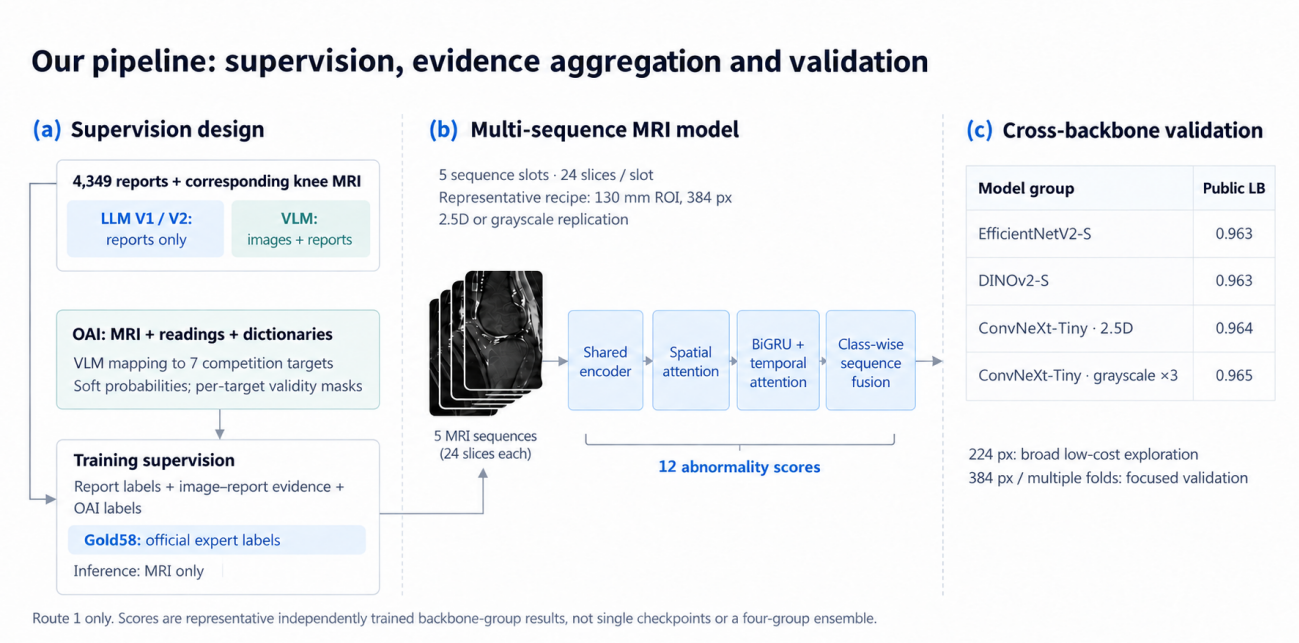


Figure 1. Path-one overview: supervision design, multi-sequence imaging, and cross-backbone validation. RSNA and OAI are separate supervision sources; the LLM reads reports only. The right-hand scores are representative Public LB results for independently trained backbone groups at different stages, not individual

checkpoints, a four-group ensemble, or the final team result. This figure excludes Max's CoAtNet and the R7 branch. MRI thumbnails are illustrative.

2.1 Methodology: Broad Exploration, Focused Validation

We used substantial aggregate compute to explore many inexpensive experiments in parallel, then concentrated resources on a small number of expensive validations. Early experiments mainly used **224 px inputs** to screen input representations, evidence aggregation, and supervision strategies. Promising recipes were then tested at **384 px and across multiple folds**, and transferred to EfficientNet, ConvNeXt, and DINO backbones. These backbones followed the same evolving technical path.

The role of 224 px was to screen directions; the role of 384 px was to confirm gains. We did not assume that a change would produce the same improvement at higher resolution. Data quality meant more than clear images or complete files: it also included consistent anatomical ordering, supervision aligned with target definitions, and a distinction between uncertainty and missing evidence. When more external data, models, or distillation failed to provide consistent gains, we retained the recipes supported by validation.

2.2 Supervision Design: From Report Extraction to Image–Report Alignment

Our supervision evolved through **LLM report soft labels → blending two LLM versions → VLM image–report checking and blending → OAI label mapping**. Each step addressed information or definition gaps that remained in the preceding stage.

Report-based LLM prompts in this branch used **Gemma 4B, Opus 5, and ChatGPT 6 Astra**. VLM prompts used **ChatGPT 6.1 Sol and Opus 5.5** for RSNA image–report checking. **OAI label mapping used a local Gemma 3 4B model.** This lists the scope of model usage; it does not mean that every case was labeled by all models. V1 and V2 below denote prompt and label versions, not a one-to-one assignment to particular models.

First, preserve report uncertainty. Suspicious, mild, and borderline findings are difficult to reliably reduce to 0 or 1. The first LLM version extracted findings, location, severity, negation scope, and supporting evidence, distinguishing unmentioned findings from explicit negative and positive statements. It read reports only, without MRI input. Soft scores preserved these semantic distinctions and reduced information loss from forcing ambiguous language into hard targets. They were not assumed to be calibrated probabilities.

Next, distinguish severity from the probability of meeting the positive criteria. The second LLM version more directly estimated the probability that an imaging expert would assign a positive label under the competition definitions, rather than only grading severity. The two prompts retained different perspectives, and their mixtures were compared through downstream training. In the corresponding experiment, blending the two LLM versions improved **Public AUC by about 0.001**.

VLM evidence addressed limitations of report-only supervision. Even faithful report extraction inherits report omissions and differences between clinical descriptions and the competition's imaging criteria. The image–report VLM checked evidence using **MRI from the same examination, the original report, and the provided target annotation criteria**. It assessed whether visible findings met the competition definitions. When images and reports conflicted, the saved prompts directed the model to examine image evidence rather than force agreement. Images were introduced to improve supervision–task alignment.

We retained both LLM and VLM sources. Reports summarize a clinician's review of the full examination, while the VLM contributes evidence from sampled images. Reports may omit findings; sampled images may miss relevant pathology; model readings themselves are uncertain. We therefore treated neither source as absolute

ground truth. We preserved complementary information and selected mixtures through downstream model results. Representative blending experiments improved **Public AUC by about 0.002–0.003**: the 224 px VLM-only recipe rose from **0.951 to 0.953**, and an earlier 384 px EfficientNet recipe from **0.957 to 0.960**. These are downstream comparisons at different stages, not measurements of label accuracy.

One mature recipe was:

`L = 0.7 × LLM_V1 + 0.3 × LLM_V2`
`Y = 0.7 × VLM + 0.3 × L = 0.70 × VLM + 0.21 × V1 + 0.09 × V2`

Different models also explored source and mixture variations. The new Conv model used `0.7 × Opus VLM + 0.3 × L`, while the new DINO model used `0.5 × Opus VLM + 0.5 × L`. Targets for Gold58 were finally overwritten with official expert labels. These are **training-target mixing coefficients**, distinct from RSNA/OAI sampling ratios or final ensemble weights. Already blended labels must not be mixed again as raw sources.

Supervision stage	Main issue addressed	Representative result or validation
LLM V1 soft labels	Structured representation of negation, location, severity, and borderline descriptions.	Starting supervision; no isolated estimate of improvement over hard labels.
Two-version LLM blend	Confusion between severity and the probability of a positive competition label; interpretation bias from a single prompt.	About **+0.001** in the corresponding experiment.
VLM/LLM image–report blend	Report omissions, image–report conflicts, and target-definition differences, while retaining both information sources.	About **+0.002–0.003** in corresponding experiments. The 224/384 comparisons above illustrate this gain and must not be counted again.
OAI label mapping	External grading systems are not equivalent to competition positive-label definitions.	About **+0.002–0.003** over the original rule-based mapping in our experimental records; details below.

These gains came from different stages, some involving recipe changes, and cannot simply be added. We assessed supervision quality using frozen label snapshots and downstream performance, rather than relying only on plausible textual explanations.

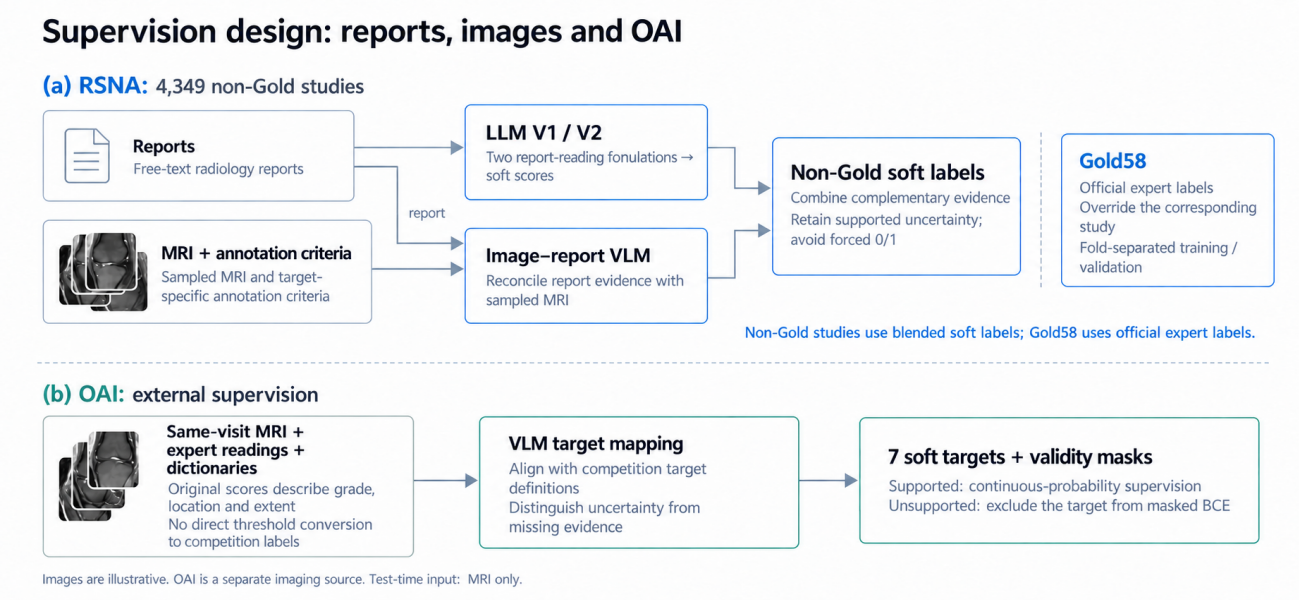


Figure 2. RSNA and OAI supervision. Report LLMs read text only; the image–report VLM receives reports, sampled MRI, and target annotation criteria. Non-Gold studies use blended soft labels, while Gold58 uses official expert labels, with training and validation separated by fold. OAI's seven soft targets are mapped using same-visit MRI, expert readings, and dictionaries. Validity masks apply to target-wise masked BCE; they do not imply that every historical auxiliary loss used the same mask. MRI thumbnails are schematic, and OAI is a separate imaging source.

2.3 OAI: Align Definitions Before Increasing External Exposure

Our OAI experiments first exposed a limitation: **detailed original scores do not automatically translate into competition labels.** OAI expert readings describe location, extent, and severity; the competition asks whether a specific abnormality meets its positive criteria. The original threshold-based rule mapping performed poorly. With misaligned definitions, simply adding more external examinations can amplify supervision bias.

Evan subsequently developed label mapping using **MRI from the same visit, original expert readings, and field dictionaries**. This produced **seven soft targets and target-wise validity masks**: medial and lateral meniscus, medial and lateral OA, PF OA, effusion, and synovitis. Relative to the original rule mapping, the experimental records showed approximately **0.002–0.003 Public AUC** improvement.

We distinguished **supported but ambiguous evidence** from **insufficient evidence**. The former retained a continuous soft score; the latter was controlled through validity flags in BCE supervision rather than filled with a negative label. The complete 12-class RSNA soft-score table, OAI target-validity masks, and image-slot presence masks represent different information.

The OAI candidate pool contained **2,399 baseline examinations and 6,250 follow-up examinations**, not 8,649 independent participants. A representative training epoch sampled approximately **3,305 RSNA studies and 2,393 OAI examinations**, or roughly **58%/42%**. This is epoch-level sample exposure, not a fixed batch composition or a label-mixing coefficient. Baseline and follow-up examinations can share participants, so candidate count is not independent information count. The three OAI sequences filled the first three sagittal, coronal, and axial slots; the remaining slots were missing. Label validity controlled which targets could be supervised.

In one evolving 384 px EfficientNet branch, blended labels, baseline OAI, and the expanded seven-target recipe reached Public scores of **0.960, 0.962, and 0.963**, respectively. These comparisons may jointly involve mapping, target scope, and candidate-pool changes, so they must not be added to the mapping gain above. More follow-up data or four-versus-seven-target supervision did not consistently help at every stage. Our conclusion was: **validate what external supervision means before deciding how much to add.**

2.4 Inputs and Architecture: Hierarchical Evidence Aggregation

Original DICOM series were assigned to five slots by plane and sequence type: `SAG_FLUID_FS`, `COR_FLUID_FS`, `AX_FLUID_FS`, `COR_T1`, and `SAG_T1`. Preprocessing included physical-position sorting, central-depth sampling, physical-size center cropping, intensity normalization, left/right orientation harmonization, and missing-slot masks. The mature 384 px pipeline typically cached **24 slices per slot with a 130 mm center crop**, joining shards by study ID.

Two input representations appeared in successful model groups. **2.5D** stacked three adjacent slices as three channels, creating 11 overlapping windows from 24 slices at stride 2. **Grayscale replication** copied each single slice into three channels, retaining 24 slice tokens per slot. Each checkpoint must use its own training-time representation. The DINOv2 and RadImageNet branches used their corresponding 392 px and 224 px inputs.

Most models shared the following evidence-aggregation path: **shared visual encoder → label-wise spatial attention → within-sequence BiGRU and temporal attention → class-wise fusion across five sequences → 12 outputs**. Missing slots were excluded from encoding, auxiliary BCE, and fusion.

The architecture organized evidence at three levels. At the **spatial level**, class-specific aggregation sought relevant regions, reducing dilution of local findings by background under global averaging. At the **slice level**, a per-sequence BiGRU retained context, followed by label-wise temporal attention over relevant slices. At the **sequence level**, learned weights represented different findings' preferences for planes and contrasts. Fusion used global class-wise weights, renormalized over present slots, rather than case-dependent gating.

The shared encoder reused visual features, while each sequence retained its aggregation branch to accommodate contrast and plane differences. Label-wise aggregation could learn evidence weights from examination-level supervision without additional lesion boxes. Attention weights themselves were not validated lesion localization. Architecture iteration used ChatGPT and Claude for discussion and implementation, followed by code checks and experimental selection. EfficientNet, ConvNeXt, and DINO backbones carried the same evolving recipe, producing complementary input and label variants.

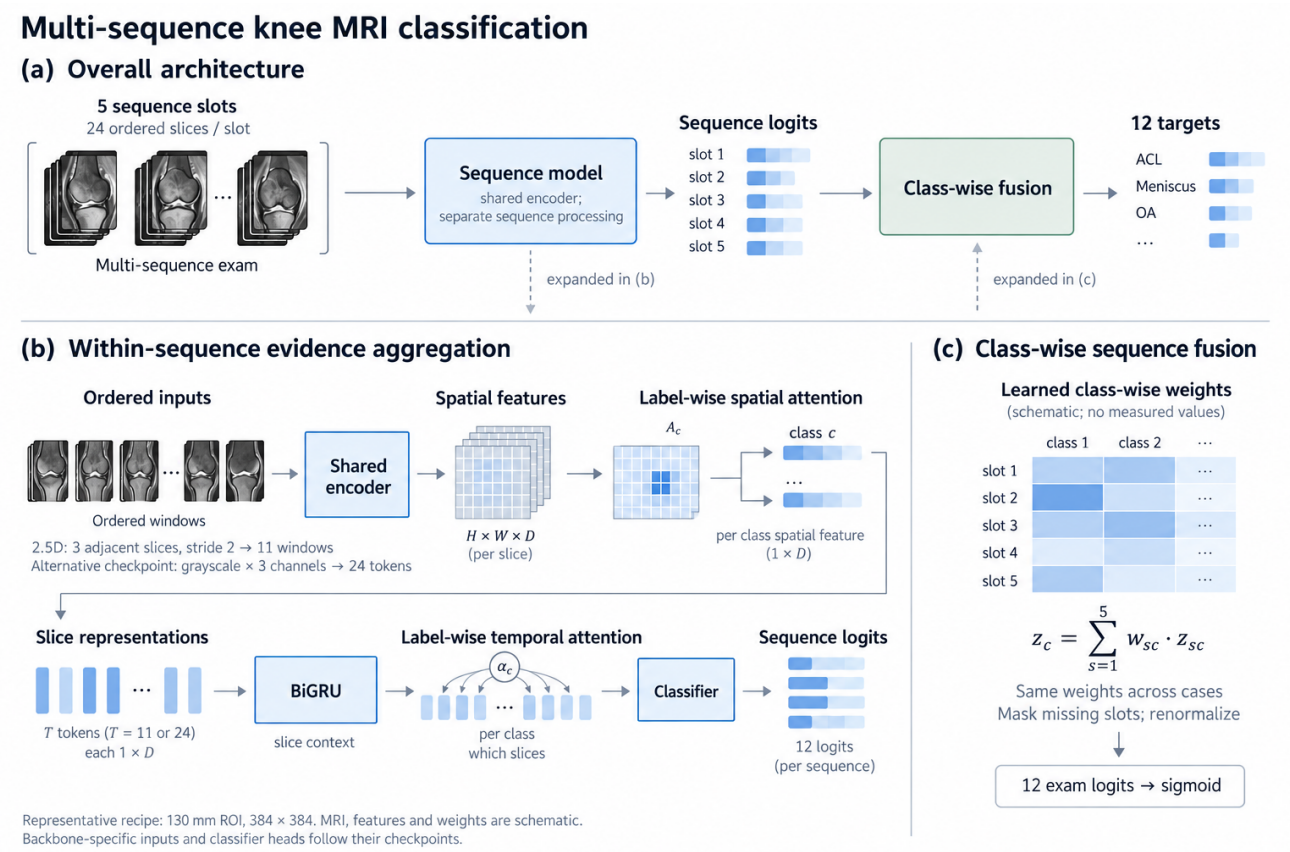


Figure 3. Hierarchical aggregation of multi-sequence MRI evidence. (a) Sequences are processed separately with shared encoder parameters, followed by class-wise fusion of sequence-level classification results. (b) Label-wise spatial aggregation, BiGRU slice-context modeling, and label-wise temporal aggregation. (c) Global class-wise weights, shared across cases, fuse sequence logits; missing sequences are masked and weights renormalized. 2.5D and grayscale replication are alternative checkpoint-specific inputs. MRI, features, and weight matrices are schematic, not actual cases, lesion localizations, or measured weights. Exact input sizes and classifier heads follow the training code.

2.5 Training Recipe and Main Improvements

The mature multi-sequence ConvNeXt recipe used AdamW, bf16, EMA, batch size 4 with four gradient-accumulation steps, a representative uniform learning rate of `2e-4`, and 12 epochs. DINO and other backbones retained appropriate learning rates and freezing settings for their initialization. Geometric transforms were shared across slices within a study, and intensity augmentation was consistent within each sequence. EMA weights were evaluated and checkpointed. Auxiliary expert-head BCE was computed only for present slots.

Category	Purpose	Representative experimental result
Effective inputs and anatomical consistency	Physical sorting, consistent orientation, tighter crops, and higher resolution to retain joint detail.	EfficientNet changed from 224 px/crop160 to 320 px/crop130: Public **0.925 → 0.937**. Crop and resolution changed together.
Hierarchical evidence selection	Class-specific spatial and slice selection, followed by sequence fusion.	224 px attention recipe: **0.939 → 0.943**, about **+0.004**.

Category	Purpose	Representative experimental result
Slice coverage and overlapping windows	Expose more evidence and reduce discontinuities at window boundaries.	Corresponding coverage changes: about **+0.001**; overlapping-window recipe: about **+0.003**.
Robust training	Address geometric, intensity, and acquisition variation.	Early geometric augmentation: **0.937 → 0.942**; an intensity-augmentation recipe: about **+0.003**.
Optimization and stability	EMA stabilizes weights; auxiliary BCE and ranking losses add training constraints.	Experiment notes estimated about **+0.002** for EMA and **+0.001** for ranking; these were not equally strict isolated ablations.
Supervision design	Two LLM versions, image–report blending, and OAI target mapping.	See the supervision table; do not count these again as separate gains from adding OAI or other recipe changes.
Limited TTA	Average deterministic views while balancing gains and cost.	Later three-angle comparisons improved Public and Private by about **+0.001**; the earlier **+0.003** estimate used a more expensive nine-view setting.

The table combines representative comparisons from different stages, not additive gains. Physical sorting improved the corresponding Public result by about 0.001 without a matching Private gain, reminding us to distinguish sound motivation from measured generalization. Low-resolution findings required high-resolution confirmation: one 24-slice grayscale branch improved from Public/Private **0.951/0.936** at 224 px to **0.956/0.943** at 384 px. This supported higher-resolution validation, but did not establish equal transfer gains for every trick.

2.6 Milestones and Directions Without Consistent Gains

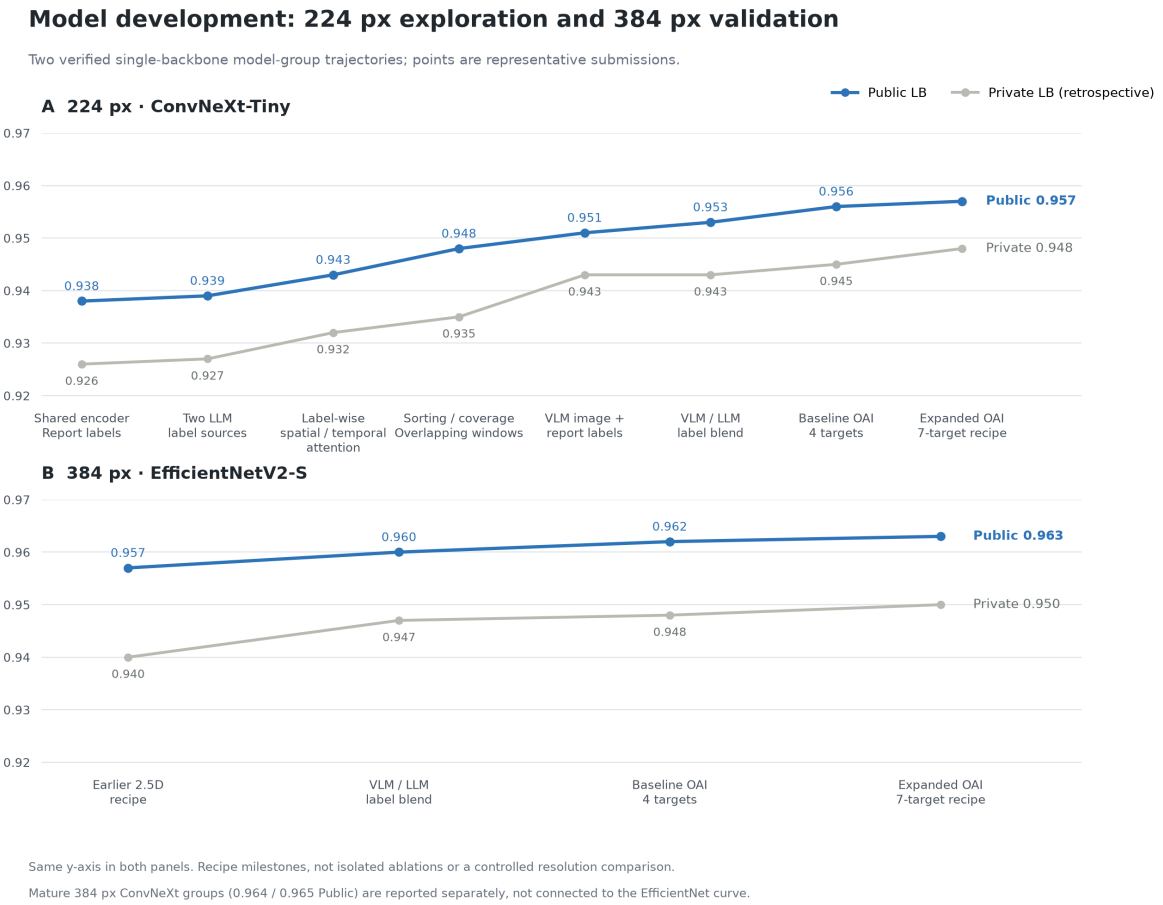


Figure 4. Two development trajectories: 224 px ConvNeXt-Tiny above and 384 px EfficientNetV2-S below. Blue denotes Public LB; gray denotes retrospective Private LB. Both panels use the same y-axis. Lines connect representative submissions within each trajectory, not a controlled resolution comparison or isolated ablations. The 224 px trajectory runs from 0.938/0.926 to 0.957/0.948; the 384 px EfficientNet trajectory runs from 0.957/0.940 to 0.963/0.950. Per-point submission links are provided in the Figure 4 data record.

These are selected milestones, not all experiments or a claim that every change helped. For example, another early LLM-blend branch scored 0.937/0.925, below its 0.938/0.926 baseline. Later successful mixtures do not imply that blending invariably improves results. Mature 384 px ConvNeXt results of 0.964/0.965 are reported separately below, not connected to the EfficientNet curve.

After transferring the mature recipe, the 2.5D and grayscale-replication ConvNeXt-Tiny groups reached Public scores of 0.964 and 0.965; EfficientNetV2-S and DINOv2-S reached 0.963. These were generally multi-fold model groups, not single checkpoints. The following results represent different development stages, not a uniform comparison of final ensemble components.

Representative model group	Public LB	Private LB
ConvNeXt-Tiny 384, 2.5D	0.964	0.954
ConvNeXt-Tiny 384, grayscale replication	0.965	0.954
EfficientNetV2-S 384	0.963	0.950

Representative model group	Public LB	Private LB
DINOv2-S, 392 px input	0.963	0.950
TinyViT-21M 384	0.961	0.949
RadImageNet ResNet50, 224 px input	0.957	0.943

Earlier backbones such as TinyViT and RadImageNet belonged to the same development path, but not all entered the final four-group multi-backbone branch.

Within our branch, expanding OAI, changing supervised targets, adding teacher predictions, and increasing model count did not consistently help across stages. Four-versus-seven-target results depended on the candidate pool and label version; distillation also depended on teacher quality, target alignment, and validation isolation. We retain these negative results to explain the evidence behind our choices. They do not negate the CoAtNet branch's distillation gains under a different supervision pipeline.

3. Path Two: Iterative Label Distillation and CoAtNet MIL

Max primarily drove iterative label distillation and consensus-label construction, CoAtNet-2 training, and all-window fine-tuning. Max also refined ensembling and post-processing strategies, providing complementary model branches and integration support for the final team solution. We describe this branch's labels, imaging inputs, and training separately; its splits and metrics should not be conflated with those in Section 2.

What mattered most in this path. Labels were improved in rounds: in most rounds, predictions from earlier models on studies they had not trained on (out-of-fold, or OOF, predictions) went into the next labels. The first two results below compare whole rounds, so they are not strict single-change tests.

- **GPT-6 Astra readings of the MRI images, without reports,** mixed into the labels: the largest gain of any one round, about **+0.019 Public and +0.017 Private** (checkpoint selection also changed in that round; Section 3.1).
- **The label rounds overall:** from **0.924 to 0.964 Public**, including the step above. The later rounds' labels also drew on R7's predictions and on the multi-backbone path's VLM/LLM labels and model predictions (Section 4.1), and the backbone, image preparation and training data changed along the way (Figure 6).
- **A short final fine-tuning on every window:** the four full-data CoAtNet-2 models were each trained with a different random seed on all training studies except 100 kept aside for choosing epochs, then given four extra epochs that showed them every three-slice window from the study's retained sequences instead of 12 random ones (Section 3.4). Tested on four-fold versions of these models (the full-data models have no out-of-fold predictions), this gained about **+0.0021 OOF** and **+0.0006** in the team ensemble.

In this section, public-leaderboard steps of about 0.001 are treated as within chance: the public part has only about 390 studies, and scores are shown to three decimals. The first two gains above are far larger, and the third was measured out of fold.

3.1 Label Iteration: Reports, Independent Image Readings, and OOF Predictions

Max's distillation path began with public Raptor report soft labels, added two other report-reading sources, and then incorporated **GPT-6 Astra readings of MRI images without reports**. These image-only readings supplied a different source of evidence from the image–report VLM. Each round's models then produced out-

of-fold predictions on studies they had not trained on. Predictions were mapped back to the label scale and blended with report and image-reading scores for the next training round. Each new model started again from ImageNet weights; training used neither an online teacher network nor an additional temperature-based distillation loss.

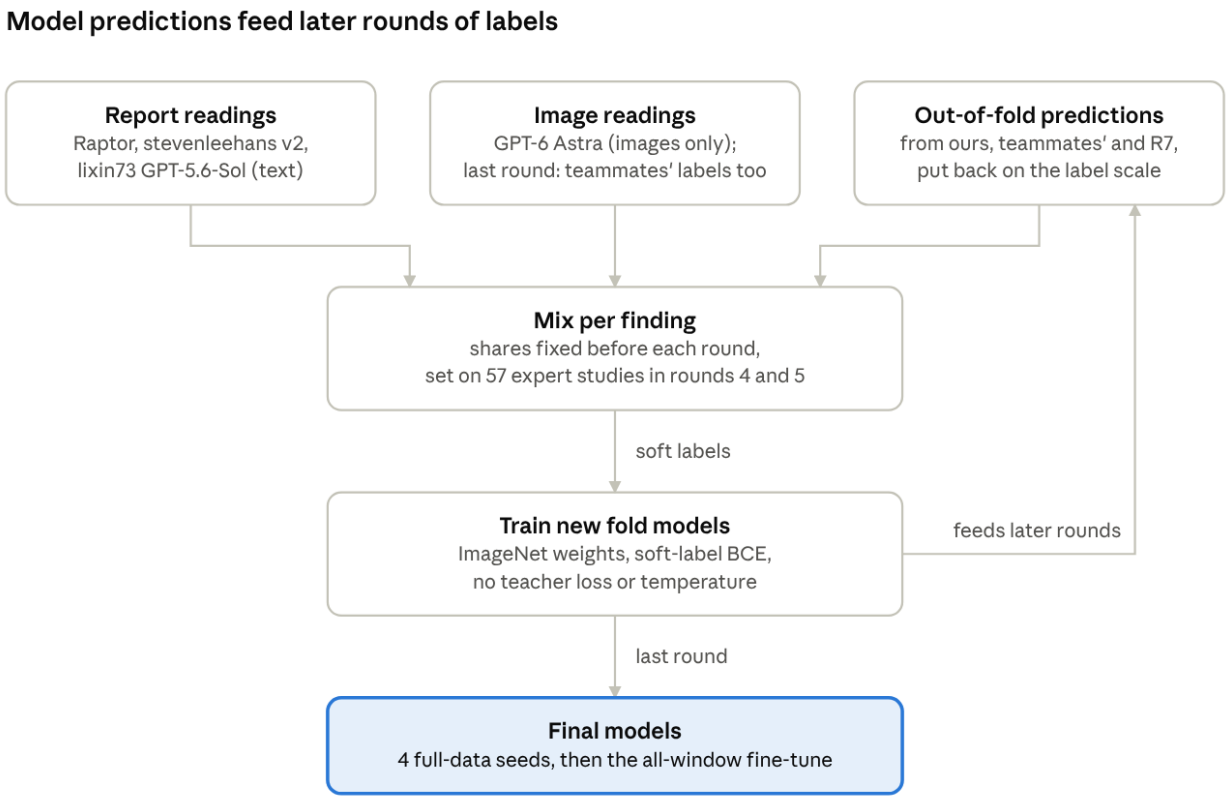


Figure 5. Label-distillation workflow. In the original figure, “ours” denotes the CoAtNet branch and “teammates” denotes the multi-backbone imaging branch.

The fifth-round distilled labels were called **v7**. Later consensus labels combined supervision and predictions from both team paths, as detailed in Section 4.1.

This path improved from **0.924 Public** with report labels to **0.964 Public** with full-data seeds. The round that introduced GPT image readings improved Public and Private by approximately **0.019** and **0.017**, respectively. Checkpoint selection also changed at that stage, so these are round-level results rather than strict single-variable ablations.

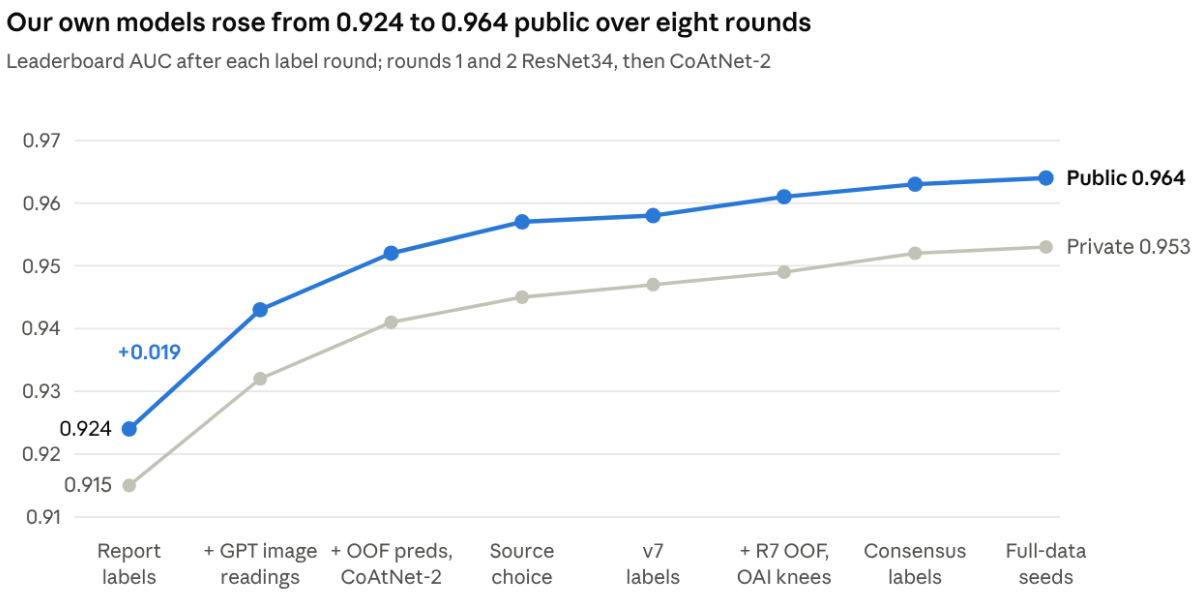


Figure 6. Results across distillation rounds. Labels, backbone, or cache settings also changed between some rounds.

3.2 Inputs and Architecture: Window-Level MIL

The window-level MIL branch used the approximately **73M-parameter** `coatnet_rmlp_2_rw_384.sw_in12k_ft_in1k`, adapting the public Raptor preprocessing and attention-MIL head. Its five slots comprised sagittal fluid-sensitive/other, coronal fluid-sensitive/other, and axial sequences. After physical-position sorting, it retained 6%–94% of the depth, used a **140 mm** center crop, and cached images at **336 px** before resizing to **384 px**. Three adjacent cached slices formed an input window, with up to approximately 134 windows per study.

The branch computed class scores for individual windows and used class-wise attention MIL to aggregate windows across the study. Windows from all five sequences entered one bag, without additional plane or positional encoding in the head. This provided a different evidence-aggregation approach from within-sequence BiGRU modeling followed by class-wise sequence fusion.

3.3 OAI: Baseline Knees as External Supervision

The CoAtNet branch used 2,398 baseline OAI knees and no follow-up visits. Fixed rules first mapped MOAKS (MRI Osteoarthritis Knee Score) grades to the competition findings. For the seven findings both paths cover (both menisci, the three OA findings, effusion and synovitis), these values were averaged 50/50 with the multi-backbone path's OAI labels (Section 2.3), in which a model read each knee's MRI together with its MOAKS grades. ACL and Baker's cyst kept the rule-based value; MCL, contusion and fracture were left blank. In two experiments in this branch, both run before the 50/50 averaging and using only the rule-based labels, adding the OAI knees improved Public by about 0.001 each time (0.957 → 0.958 for this branch's models alone, and 0.962 → 0.963 in a blend with R7). Steps this small are within chance on the public leaderboard, so these two tests alone cannot show how much OAI helped this branch.

3.4 Training, Full-Data Seeds, and All-Window Fine-Tuning

Regular CoAtNet training used soft-label BCE, fixed class-specific positive weights, AdamW, and a one-cycle schedule, with peak learning rates of `3e-5` for the encoder and `1e-3` for the head. It used effective batch size 8, **24 epochs**, and **12 randomly sampled windows per study**. Each of the final four full-data seeds (42–45) then received **four epochs of all-window fine-tuning**: all windows from retained sequences were used per training study, learning rates were reduced, and BatchNorm statistics were frozen. Weights from three selected epochs were averaged. In a four-fold consensus comparison, all-window fine-tuning improved **OOF by about 0.0021** and the team ensemble by approximately **0.0006**.

At inference, CoAtNet uniformly sampled **56 windows per study** from the full set. “56” is the number of prediction windows per examination, not the number of models. This inference sampling differs from both regular training with 12 random windows and all-window fine-tuning.

4. Team Collaboration: Shared Supervision and Complementary Ensembling

The two paths differed in supervision, input organization, and evidence aggregation. Later collaboration occurred at two levels: combining labels and OOF predictions into consensus training targets, and ensembling trained model groups at inference. These were distinct mixing operations.

Design	Multi-backbone, multi-sequence group	CoAtNet-2 branch
Slice inputs	11 overlapping 2.5D windows or 24 grayscale-replicated tokens per slot.	Three-slice windows across a study; regular training randomly sampled 12 per study.
Spatial and depth aggregation	Label-wise spatial attention, BiGRU, and temporal attention.	Class-wise attention MIL over window features.
Sequence handling	Class-wise fusion of five standard slots, with missing-slot masking.	Windows from five sequences enter one bag.
Typical image coverage	130 mm, 384 px; some backbones use different input sizes.	140 mm, cached at 336 px and resized to 384 px.

4.1 Later Consensus Labels: Where the Paths Converged

The fifth-round CoAtNet distilled labels were **v7**. Final consensus labels combined three sources: v7, VLM/LLM labels, and OOF predictions from multi-backbone models, CoAtNet models, and R7. The prediction source was first blended by class-wise ranks, then mapped back to the soft-label range. Its internal weights were **0.45** for multi-backbone models, **0.30** for CoAtNet, and **0.25** for R7. Gold58 retained official targets; OAI used separate targets and masks. Predictions from the resulting models were therefore not statistically independent, and OOF agreement against shared teacher-derived targets must not be treated as an independent test result.

The 0.45/0.30/0.25 coefficients describe the internal OOF-prediction mixture used to construct consensus labels. They are neither final submission ensemble weights nor the LLM/VLM recipe from Path One.

4.2 Model Groups, Fusion, and Results

The multi-backbone branch contained four groups: **DINOv2-Small, EfficientNetV2-S, DINOv3-distilled ConvNeXt-Tiny, and ConvNeXt-Tiny**. Folds, seeds, and test views were first aggregated within each group, followed by ensembling with four full-data CoAtNet-2 seeds and the public R7 branch.

The final team ensemble averaged the four imaging groups' probabilities equally, then used branch weights of **70% multi-backbone, 21% CoAtNet, and 9% R7**. Per-class predictions were converted to percentile ranks and combined with a noisy-OR-style operation. The Lateral OA, PF OA, and Synovitis ranking scores were then reduced by `0.10 × Medial OA`, `0.10 × Medial OA`, and `0.20 × Effusion`, respectively, before outputting final per-class ranks. This post-processing changed study ordering; it was not probability calibration.

A separate team submission with **50% multi-backbone, 30% CoAtNet, and 20% R7** scored **0.967 Public/ 0.956 Private**. The final selected team submission used **70% multi-backbone, 21% CoAtNet, and 9% R7** and scored **0.967 Public/0.957 Private**, placing 10th.

In the 50/30/20 submission, the post-processing worked as follows:

1. **Ranks.** For each finding, each branch's predictions became percentile ranks r (between 0 and 1) among the m studies it scored.
2. **Noisy-OR.** Each rank became −log(1 − r + 0.5/m), then the branches were averaged with their weights. This value stays small for most studies and grows fast near the top, so a study that any one branch ranks near the top rises, as if that branch cast an independent "present" vote.
3. **Bell-curve scores.** The blended column was ranked again, and each rank was turned into the matching value of a standard normal (bell-curve) distribution.
4. **Corrections.** The three subtractions above were applied to those values. The idea: part of the Lateral OA, PF OA and Synovitis scores is a general "arthritic knee" or "fluid" signal shared with Medial OA or Effusion, and the subtraction removes some of it. The same subtraction with clinically unrelated partner findings gave no gain.
5. **Output.** The percentile rank of the result.

This was tested out of fold on 5 random 50/50 splits of the 4,296 validation studies, scoring each half separately, with four-fold CoAtNet-2 models standing in for the four full-data models (which have no out-of-fold predictions). With the subtraction sizes chosen on one half and scored on the other, steps 2–4 gained about **+0.0004** over a plain weighted average of ranks, positive in all 10 halves. On the leaderboard, an earlier version of that submission scored 0.966 Public / 0.955 Private both with and without this step, as expected for a gain this small.

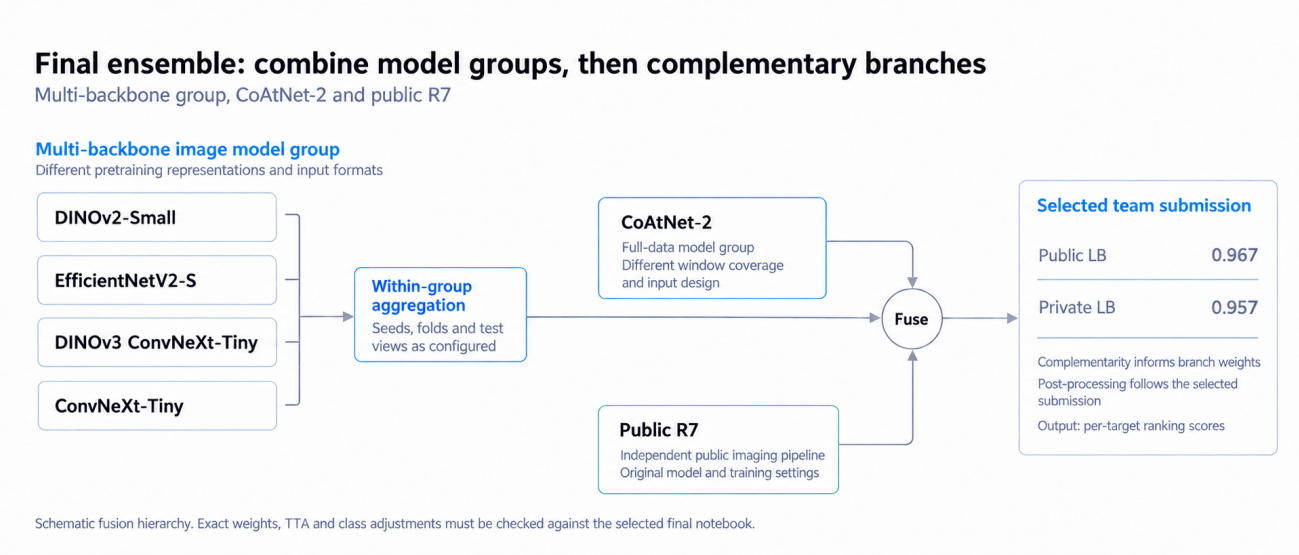


Figure 7. Final ensemble: 70% multi-backbone group, 21% CoAtNet-2, and 9% public R7. The team's selected submission scored Public 0.967 and Private 0.957.

Stage	Public LB	Private LB
Mature ConvNeXt-Tiny group, 224 px → 384 px	0.957 → 0.964	—
Later ConvNeXt-Tiny model group	Approximately 0.965	—
Four full-data CoAtNet-2 seeds, before all-window fine-tuning	0.964	0.953
Team ensemble, submission	0.967	0.956
Final selected team submission, (70/21/9)	**0.967**	**0.957**

All-window fine-tuning, 56-window inference, and dual-T4 workload allocation balanced coverage against the nine-hour submission limit. Inference also used concurrent DICOM reading, caching, and limited TTA. Each model retained its training-time input, normalization, and checkpoint settings. Neither report LLMs nor image–report VLMs ran at test time.

5. Validation, Reproduction, and Resources

5.1 Interpreting the Two Paths' Validation Results Separately

OOF, Gold58, and Public LB results from the two paths are not interchangeable measurements. Their roles are summarized below.

Item	Multi-backbone, multi-sequence path	CoAtNet path
Splits	Four folds were grouped by normalized report hash, using report groups rather than patient or hospital identifiers.	Final four-fold validation covered 4,296 studies; the 58 expert studies and 53 studies whose reports did not mention target abnormalities were always included in training.
Gold58		

Item	Multi-backbone, multi-sequence path	CoAtNet path
	Official targets; excluded from training in their assigned validation fold but included in other folds. Primarily a small-sample diagnostic.	Always included in later-stage training; expert studies also informed source-weight selection, so they were not an independent test set.
Local metrics	Derived AUC measures ranking against current report/image–report-derived labels; the Gold sample is small.	Separate labels and model-selection procedure; full-data seeds additionally held out 100 studies for fine-tuning selection.
Later label dependencies	Labels and predictions contributed to team consensus.	Consensus used predictions from multi-backbone models, CoAtNet, and R7; later results were not statistically independent.

Public LB supplied unseen-data feedback during the competition; Private LB was used only for retrospective analysis. Gains in the text span different experimental stages and cannot be added. OOF consistency under shared teacher targets is not an independent test score. Actual inference cost also matters for model selection.

5.2 Code, Models, and Data

Resource	Purpose	Link
Our multi-backbone code and labels	224/384 preprocessing, training, prompts, historical six-/eight-model inference, and seven frozen RSNA label files.	[Work in progress], [Work in progress]
RSNA Knee data	Training/test MRI, reports, and official Gold58.	Competition data
OAI	Knee MRI and MOAKS readings.	OAI; CoAtNet branch DOI: 10.15154/wfxg-g926; multi-backbone branch: NDA Collection 2343, NDA Study S3430, DOI: 10.15154/vgnc-0x45
Raptor	Initial report labels for the distillation path; public reference for imaging preprocessing and the MIL head.	Report labels, Training code
Public R7	Ensemble branch; its OOF predictions also contributed to later distilled targets.	Weights, Inference code
Other public report labels	Additional report-reading sources.	stevenleehans, lixin73, Pilkwang

The public repository's four source-label files are read and blended once by the training scripts. The three `*_labels_used.csv` files are snapshots of actual training targets and must not be reused as raw mixture inputs. Original prompts and necessary processing scripts are included. Trained weights have not yet been released; their release method will follow organizer requirements and asset permissions. OAI labels and images still require preparation under the applicable access conditions.

The public branch provides code and labels for the historical 224/384 multi-backbone pipeline. The complete CoAtNet pipeline and final team submission code are not included.

5.3 Backbone Initialization and Pretrained Weights

The table lists the public weights used to initialize our image models. Each identifier specifies a `timm` weight variant and links to its model card. Input sizes refer to our task configuration, rather than a common resolution across the original pretraining recipes.

Model group	Pretrained weight identifier	Pretraining source and method	Our input	Weight source
ConvNeXt-Tiny	`convnext_tiny.fb_in22k_ft_in1k_384`	Supervised ImageNet-22k pretraining → ImageNet-1k fine-tuning	384 × 384	Model card
EfficientNetV2-S	`tf_efficientnetv2_s.in21k_ft_in1k`	Supervised ImageNet-21k pretraining → ImageNet-1k fine-tuning	384 × 384	Model card
DINOv2-Small / ViT-S/14	`vit_small_patch14_dinov2.lvd142m`	LVD-142M; DINOv2 self-supervised pretraining	392 × 392	Model card
DINOv3 ConvNeXt-Tiny	`convnext_tiny.dinov3_lvd1689m`	LVD-1689M; official feature weights distilled from DINOv3 ViT-7B	384 × 384	Model card
CoAtNet-2	`coatnet_rmlp_2_rw_384.sw_in12k_ft_in1k`	Supervised ImageNet-12k pretraining → ImageNet-1k fine-tuning	384 × 384	Model card

These weights initialize the encoders before task training with competition supervision and each branch's OAI recipe. They are distinct from the RSNA task checkpoints loaded at inference, such as `fold0_final.pt`. Distillation in the DINOv3 row refers to the official teacher–student visual pretraining process, separately from our report-derived soft targets, OOF consensus, and iterative label distillation. ImageNet-12k is a subset of the full ImageNet-22k dataset.

The public R7 branch directly uses the authors' released knee MRI task weights, linked in Section 5.2; it is not an additional model group that we retrained from the initializations above.

6. Acknowledgements and References

6.1 Public Resources and Foundational Methods

- We thank RSNA, the expert annotators, and Kaggle for the competition and data.
- We thank the authors of the public report-label tables: dreaddevelopment, Pilkwang, stevenleehans, and lixin73. These resources supported initial supervision and source comparisons in the label-distillation path.
- We thank dreaddevelopment for the public Raptor training notebook, an implementation reference for image preprocessing and the attention-MIL head in the CoAtNet branch.
- We thank heliosli for the public R7 inference code and weights. R7 contributed to the final ensemble, and its OOF predictions also informed the later consensus labels.
- We thank timm for backbone implementations and pretrained-weight interfaces, and the DINOv2 and DINOv3 teams for their pretrained models and public work.
- We thank RadImageNet for pretrained weights used in the early ResNet50 experiments.

6.2 OAI Data Acknowledgement

We used knee MRI and expert readings from the Osteoarthritis Initiative (OAI) for external supervision. The **multi-backbone branch** uses **NDA Collection 2343; NDA Study S3430; DOI: 10.15154/vgnc-0x45**. The **CoAtNet branch** has a separate NDA Study DOI: 10.15154/wfxg-g926.

> Data and/or research tools used in the preparation of this manuscript were obtained and analyzed from the controlled access datasets distributed from the Osteoarthritis Initiative (OAI), a data repository housed within the NIMH Data Archive (NDA). OAI is a collaborative informatics system created by the National Institute of Mental Health and the National Institute of Arthritis, Musculoskeletal and Skin Diseases (NIAMS) to provide a worldwide resource to quicken the pace of biomarker identification, scientific investigation and OA drug development. Dataset identifier(s): NDA Collection 2343; NDA Study S3430; DOI: 10.15154/vgnc-0x45 (multi-backbone branch); DOI: 10.15154/wfxg-g926 (CoAtNet branch).

The OAI acknowledgement follows the official NDA wording.

6.3 Method References

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